

RF Measurements Help to Win Lease Approval

Showing how placing a cell site near a high school would reduce RF emissions from wireless handsets used by students helped to persuade a board of education to leave in effect a previously approved lease for a tower at an athletic field.

AGL report

Members of the Marshalltown (Iowa) Community School District Board of Education voted in the majority to keep in place an agreement with U.S. Cellular for a proposed cell tower that would take the place of, yet resemble, a light pole at the Marshalltown High School football field. The vote, taken on

January 5, came about as a result of concerns brought to the board by residents near the school regarding the possible effect upon health that radio-frequency emanations from the tower might have.

Although the Telecommunications Act of 1996 does not allow government agencies to take into account the health

effects of RF emitted from cell towers and other radio transmitters when making decisions about site approval, landlords may consider health effects when deciding whether to lease.

U.S. Cellular asked Marv Wessel of Global RF Solutions, Chandler, Ariz., to speak to the board members about the



Cell towers near schools sometimes raise questions about possible health risks.

possible health effects. Afterward, *AGL* asked Wessel about his presentation.

Wessel said that in addition to local activists promoting a website against the proposed site, one or more board members and residents made reference to the "BioInitiative Report: A Rationale for a Biologically-based Public Exposure Standard for Electromagnetic Fields (ELF and RF)," which is published on the web at www.bioinitiative.org.

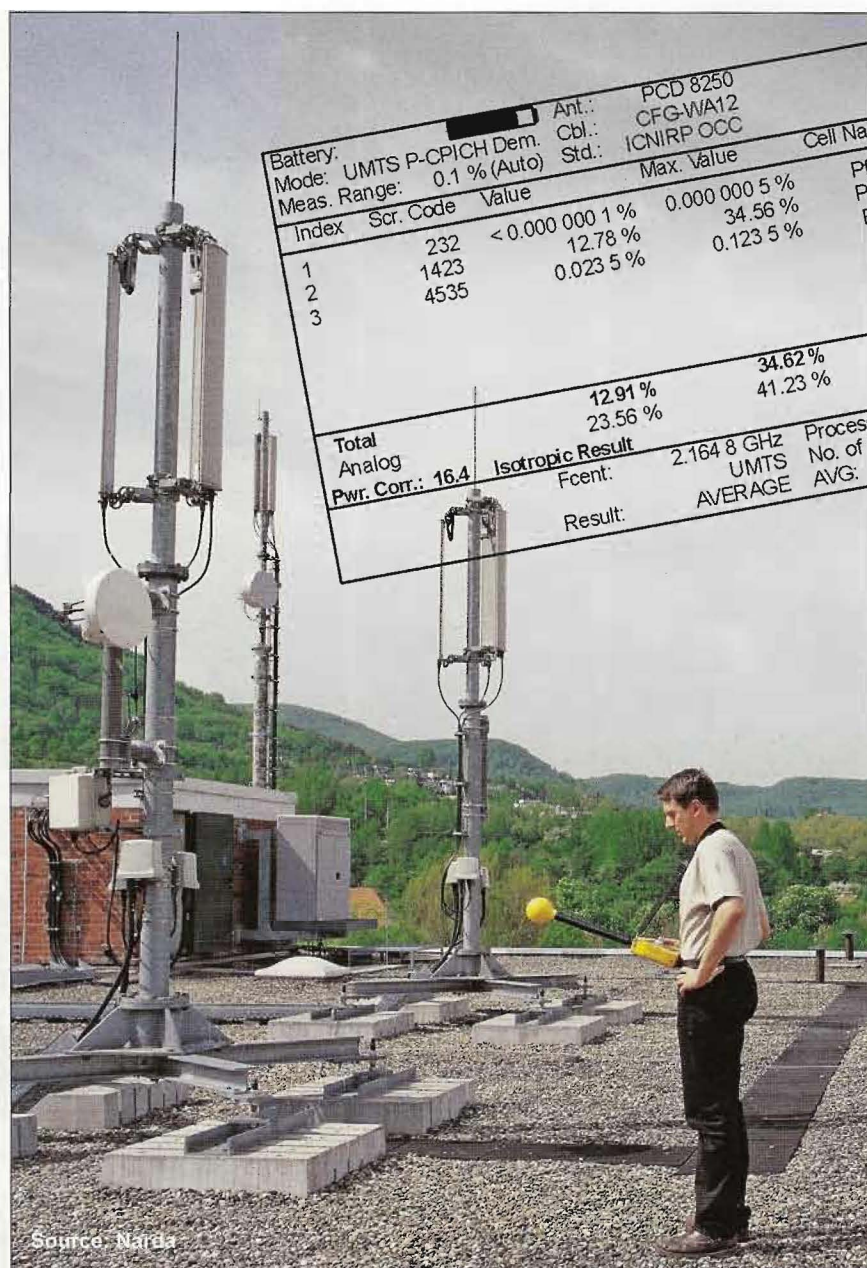
BioInitiative report

"The report claims there is a diverse range of adverse health effects related to RF emissions," Wessel said. "The BioInitiative Report seems to completely dismiss the Nationwide Cohort Study in Denmark by Christoffer Johansen, John D. Boice Jr., Joseph K. McLaughlin, and Jørgen H. Olsen that included 420,095 cell phone users and found no significant link to cell phone use and cancer.

The peculiar thing about the BioInitiative Report is that it suggests that RF exposure limits be reduced to increase the safety margin of the possibility of nonthermal effects on humans. However, it doesn't offer specifics on what these limits should be. It merely suggests that "a precautionary limit of $0.1 \mu\text{W}/\text{cm}^2$ (which is also 0.614 volts per meter) should be adopted for outdoor, cumulative RF exposure." The ironic thing, Wessel explained, is that any properly designed cellular and PCS site, which would have antennas at least 10 meters above ground level, already contributes less than what the BioInitiative Report proposes as a precautionary limit to areas surrounding these cellular and PCS sites. However, Wessel pointed out, commercial TV and FM broadcast levels typically exceed this precautionary limit in every major metropolitan location in the country. "Curiously, they choose to ignore an RF source that has been present for several decades," he said.

Better science

Wessel said that although the BioInitiative Report appears to have as its objective making the case for eliminating or reducing wireless communications, in several places the report admits



A selective radiation meter such as the Narda SRM-3000 can be used to make safety measurements for RF compliance and to provide evidence that energy levels may be many times less than the safety minimum.

there is no definitive link between cell phones and cancer. As part of preparing for his presentation, Wessel viewed a website built by one of those opposed to the tower. "He had the BioInitiative Report on his site along with other information depicting a one-sided view against cell towers."

Wessel presented the board a different perspective based on quantitative RF measurements taken with a selective

radiation meter, which consists of a spectrum analyzer and an isotropic antenna — the Narda SRM-3000. He said this meter is one of a few portable spectrum analyzers designed and calibrated specifically to make safety measurements for RF compliance.

"It has a dynamic range much greater than broadband equipment specifically designed to perform RF compliance measurements," he said. Additionally,



Source: Marv Wessel

Although a worker on a tower may come close enough to an antenna, especially FM or TV broadcast antennas, to require protection in the form of an RF suit, the inverse-square law shows that the intensity of RF from an antenna rapidly diminishes as the distance from the antenna increases. Away from the tower, energy levels rarely surpass the health limit, especially for cellular antennas. Above, Marv Wessel is shown using a selective radiation meter to take measurements.

"you can discriminate these measurements by frequency from 50 to 3000 MHz. This enables us to specifically measure each license holder's contribution to the overall RF environment."

Wessel also described the RF measurement data that was persuasive for the Marshalltown board. "They were

concerned about the RF level that would be present at the high school if they allowed the cell site to be built. Actually, the larger risk of RF exposure comes from the cellular handset itself, even though it's still greatly below the FCC limits. The inverse-square law says that 'the RF energy at a distance

from a source drops by the square of the distance from the source.'"

The RF specialist explained that at the Marshalltown school, cellular service is provided by sites a mile or two away, which causes the handsets used at the school to emit their highest RF power output. The reason is because the distant cell site controls the handset power and raises or lowers it according to the received signal (uplink) at the cell site.

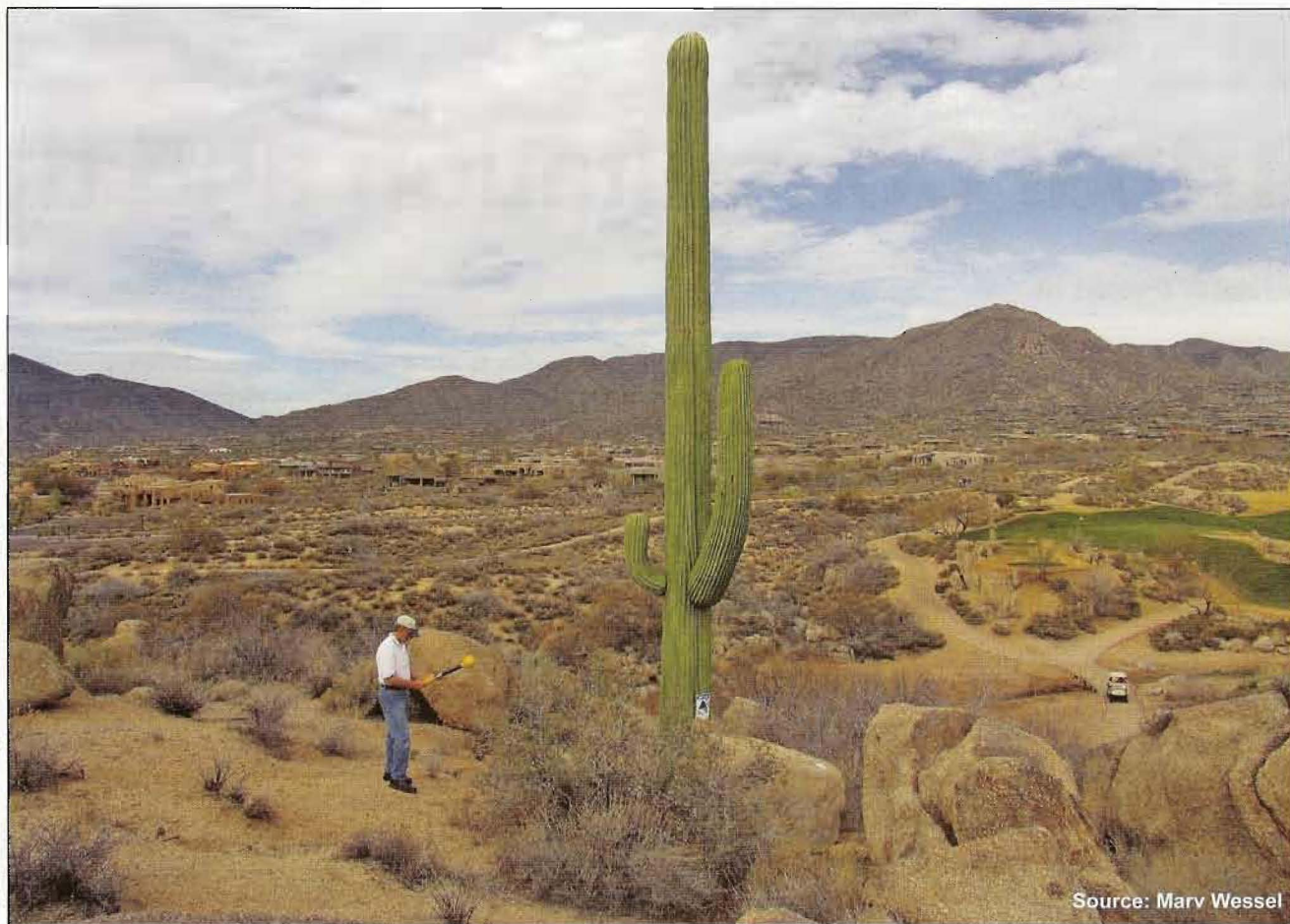
Handsets

"I took measurements at the school, and they indicated the handsets were powered up to full power during a call because of the distance to the serving cell site," Wessel said. "I told the board that it could reduce the RF exposure to students using phones by putting a site nearby because then the handsets would operate at significantly lower power. If you have a quarter-watt handset next to your head, it delivers significantly more energy to the user than does a 100-watt cell site 100 feet away and 100 feet above the ground, because of the inverse-square law."

But, Wessel said, "no one ever looks at it that way. They see the cell site as an RF radiation problem next to the school. But ironically, when you put the cell site next to a high school, where kids are huge users of handsets, the handset's power is significantly reduced by thousands

of times. A nearby site actually helps the issue."

Wessel said that one parent made an observation that students have handsets near their heads all the time. "Our measurements proved that with a nearby cell site, RF exposure from the handsets would be reduced from a level



Source: Marv Wessel

Marv Wessel is shown taking RF measurements near a cellular antenna disguised as a saguaro cactus. A person would have to scale the cactus to the antennas hidden near its top to receive unsafe RF exposure.

1,000 times (peak) below the FCC-defined safety limit to a level 1 million times below the safety limit," Wessel said. "A majority of the board members were persuaded by this approach. But some people are never satisfied."

Actual measurements

Wessel said the fact is that the handsets are going to be powered down if the cell site is nearby, and the RF exposure from the handset is the greater risk. He said that despite the evidence, some people don't want to look at the actual measurements. He said their objection sometimes is based on hysteria. "The cell sites look ominous, but the power reaching the ground in the vicinity of the site is insignificant," he said.

Wessel also described measurements he made in a New York City neighborhood where RF energy that reaches the

neighborhood from FM and TV broadcast signals is a greater percentage of the overall RF energy than the portion delivered by a cell site across the street. "They don't want to hear that, either," he said, speaking of residents who are in opposition to a neighborhood cell site. "They fear the long-term effects of RF exposure from cell sites, yet FM and TV signals have been on the air for a lot longer than cellular signals, they represent a higher percentage of the overall RF exposure than the cellular signals do, and the total is still well below the safety limit."

Common-sense approach

Wessel said the idea is to present the information in a common-sense approach. "You're never going to change the opinion of a skeptic or of the conspiracy theorists who want to believe that the

explosion in cell site growth will cause significant health risks," he said. "They call it new technology, but it's merely a new application of old technology. You can rarely change their minds. You try to present a common-sense approach to people who have become emotional by the surrounding hysteria. The cellular site opponents get people worked up and cite possible children's health effects in an effort to convince them that the cellular sites are harming the kids."

Narrowband selectivity

With measurement equipment capable of narrowband selectivity, Wessel said it is possible to select a frequency and measure the power density levels down to the picowatt per square centimeter level. "It allows you to demonstrate that cell sites are not a big deal," he said.

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